

SYSTEMS THINKING

Systems Analysis & Design

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

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Outline

1 Introduction to Systems Thinking

2 Systems Properties

3 Systems Classification



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- 1 Introduction to Systems Thinking
- 2 Systems Properties
- 3 Systems Classification



Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.

• Not all elements need to be connected to each other, but they should be connected to the system.

- The more elements there are, the more complex the system becomes. The representation must be feasible.
- Each element must have at least one connection. Isolated elements make no sense in a system.

No

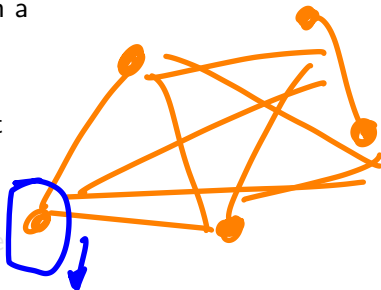


Interested



Introduction to Systems Thinking I

- A **system** is a set of interconnected elements with a common purpose.
- Not all elements need to be connected to each other, but every connection should be meaningful.
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Introduction to Systems Thinking I

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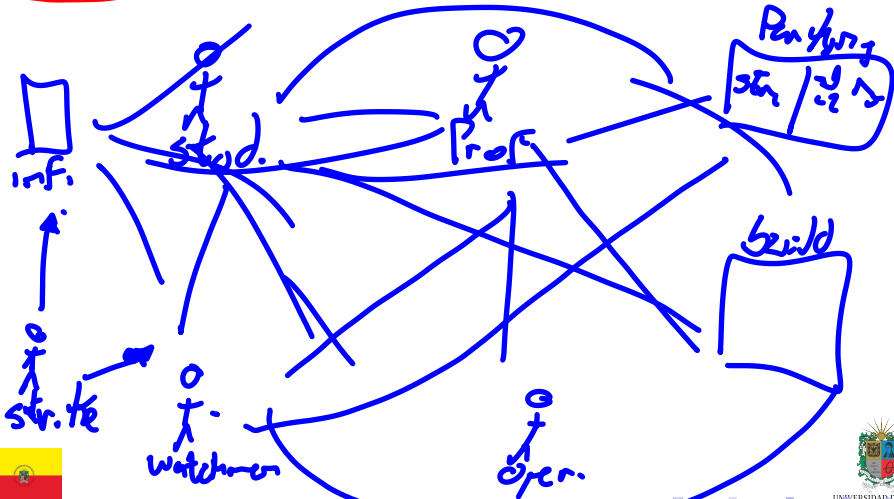
Introduction to Systems Thinking I

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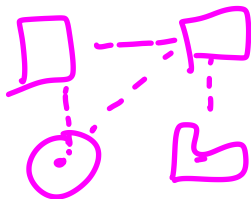


Complexity in Systems

System complexity can be defined as the number of elements and connections in a system.



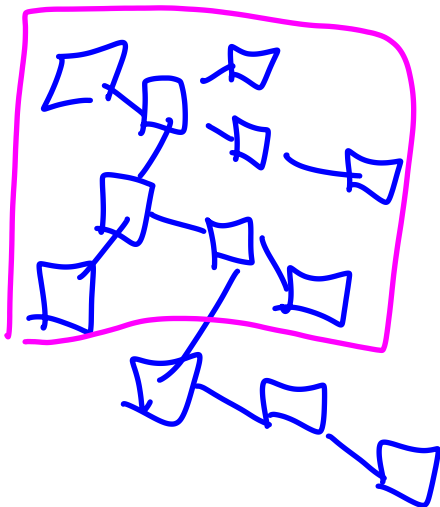
Introduction to Systems Thinking II



- In **systems thinking**, if you just **split** the parts and forget the relationships, you will **lose the full picture**.
- It is called a **holistic** approach; try to see **the whole picture** with all the meaningful details.
- **Defining the box boundaries** is sometimes tricky; as we said, not too complex, not too simple. It is like the desired universe balance of Thanos.



Introduction to Systems Thinking II



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Introduction to Systems Thinking III

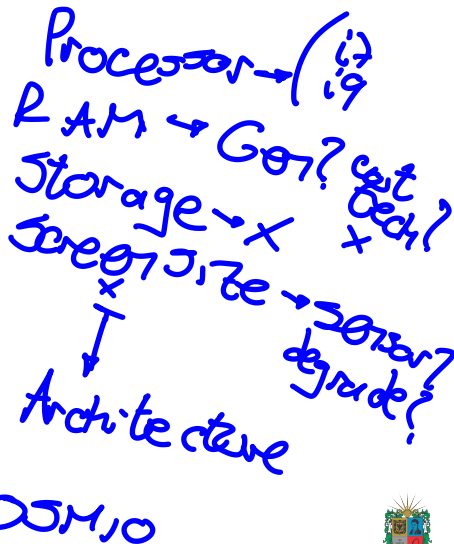
- Another important concept is **homeostasis**; it means putting a system in an equilibrium state. That is hard because systems are both not in equilibrium and resilient to change.

- A **system** is more than the sum of its parts. This means relationships, behaviors, and recovery capacity are **forgotten** when you see the system just as its parts.



Introduction to Systems Thinking III

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Introduction to Systems Thinking IV

- **Systems thinking** is a way to understand and represent problems in order to find the best possible solution.
- Thinking of a **problem as a system** lets you understand details, involved elements, and relevant information.

computation



- Systems should be visible, flexible, sustainable, provide internal feedback loops, and also look like a whole living entity.

metro - subway



Introduction to Systems Thinking IV

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2024 ~ 4 billion USD
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Cultural

OpenAI

Art. ficial Int.
Artificial Life

cybernetics



Introduction to Systems Thinking V

Automatic
calculus

Babbage

daisy
wheelmaths → algorithm → code → automation
robots

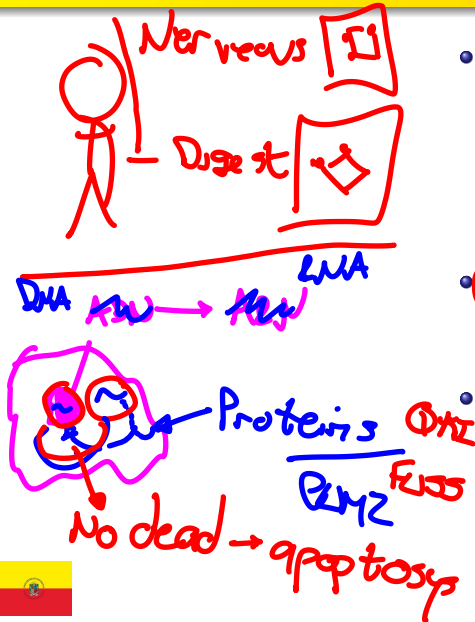
- **Computation** helps to represent behaviors mathematically. Also, it helps to find patterns and information, and simplify processes; an example of all this is Artificial Intelligence.

LLM → next word } m./10⁷
b./10⁹

- The **Bottom-Up** approach is useful when you want to see the parts and then put them together to get the full picture.



Introduction to Systems Thinking V



- **Computation** helps to **represent behaviors** mathematically. Also, it helps to **find patterns** and information, and simplify processes; an example of all this is **Artificial Intelligence**.
- **Top-Down** approach is useful when you want to see the full picture, and then split it into parts. ✓
- The **Bottom-Up** approach is useful when you want to see the parts and then connect them to get the full picture. ←



Introduction to Systems Thinking VI

- It is important to understand the **sensitivity** of the problem because it leads to better decision-making.

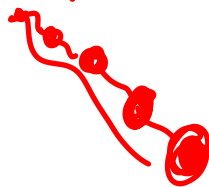
- The simplest system definition

is: given some inputs, after applying a designed process to them, you will obtain some outputs.



- In a deterministic world, the same inputs produce the same outputs. However, real life is not deterministic.

Snowball



Control



Introduction to Systems Thinking VI

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- In a **deterministic** world, the same **inputs** produce the same **outputs**. However, real life is **not deterministic**.

Engineering

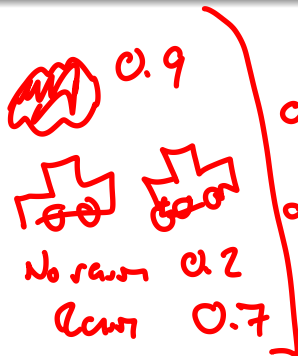
$$\begin{array}{r} I \quad , \quad O \\ 2 + 2 = 4 \\ 3 + 3 = 6 \end{array}$$

controlled system

~ TC



Introduction to Systems Thinking VII



$$0.9 \times 0.7$$

$$0.1 \times 0.2$$

Bayes Theorem

- Since **randomness** is normal in the real world, relying solely on deterministic processes is **dangerous**. Using **stochastic processes** is a better approach.
- **Stochastic processes** make use of **probability** which leads to a better representation of real-world behavior.

Here, Chaos Theory becomes a useful tool. In fact, simply, chaos can be defined as a harmonious balance between rules and randomness.



Introduction to Systems Thinking VII

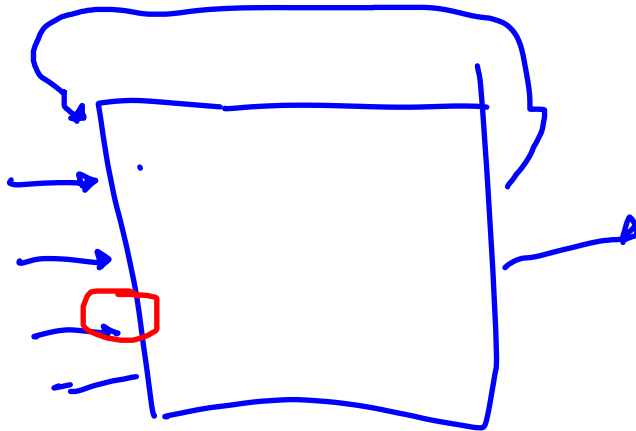


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Chaos



Systems Structure Draw



Case of Study: Transportation System



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Systems Properties I

- **Emergence** is a property of systems that means that the **whole system** is more than the **sum of its parts**.
- **Interconnectedness** is a property of systems that means that all the **elements** are **connected** in a **meaningful** way.
- **Feedback** is a property of systems that means that the system has **internal loops** that **control** the system behavior.
- **Hierarchy** is a property of systems that means that the system has **levels of organization**.



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- **Equifinality** is a property of systems that means that the system can reach the **same goal** from **different paths**.
- **Permeability** is a property of systems that means that the system can **interact** with the **environment**.
- **Dissipative** is a property of systems that means that the system can **lose energy** and **information** to the environment.
- **Homeostasis** is a property of systems that means that the system can **maintain a stable state**.



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- **Self-regulation** is a property of systems that means that the system can **regulate** itself.



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Basic Concepts

A **system** can be **classified** according to different **criteria** like **openness**, **adaptability**, **determinism**, and **linearity**.



Systems Classification I

- **Open systems** are systems that can **interact** with their **environment**.
- **Closed systems** are systems that **cannot interact** with their **environment**.
- **Adaptive systems** are systems that can **change** and **adapt** to their environment.
- **Non-adaptive systems** are systems that **cannot change** or **adapt** to their environment.



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Systems Classification II

- **Deterministic systems** are systems that produce the **same output** for the **same input**.
- **Stochastic systems** are systems that produce **different outputs** for the **same input**.
- **Linear systems** are systems that produce **proportional outputs** for **proportional inputs**.
- **Non-linear systems** are systems that produce **non-proportional outputs** for any given inputs.



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Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/systems-analysis>

